

Status report on P-Theta work

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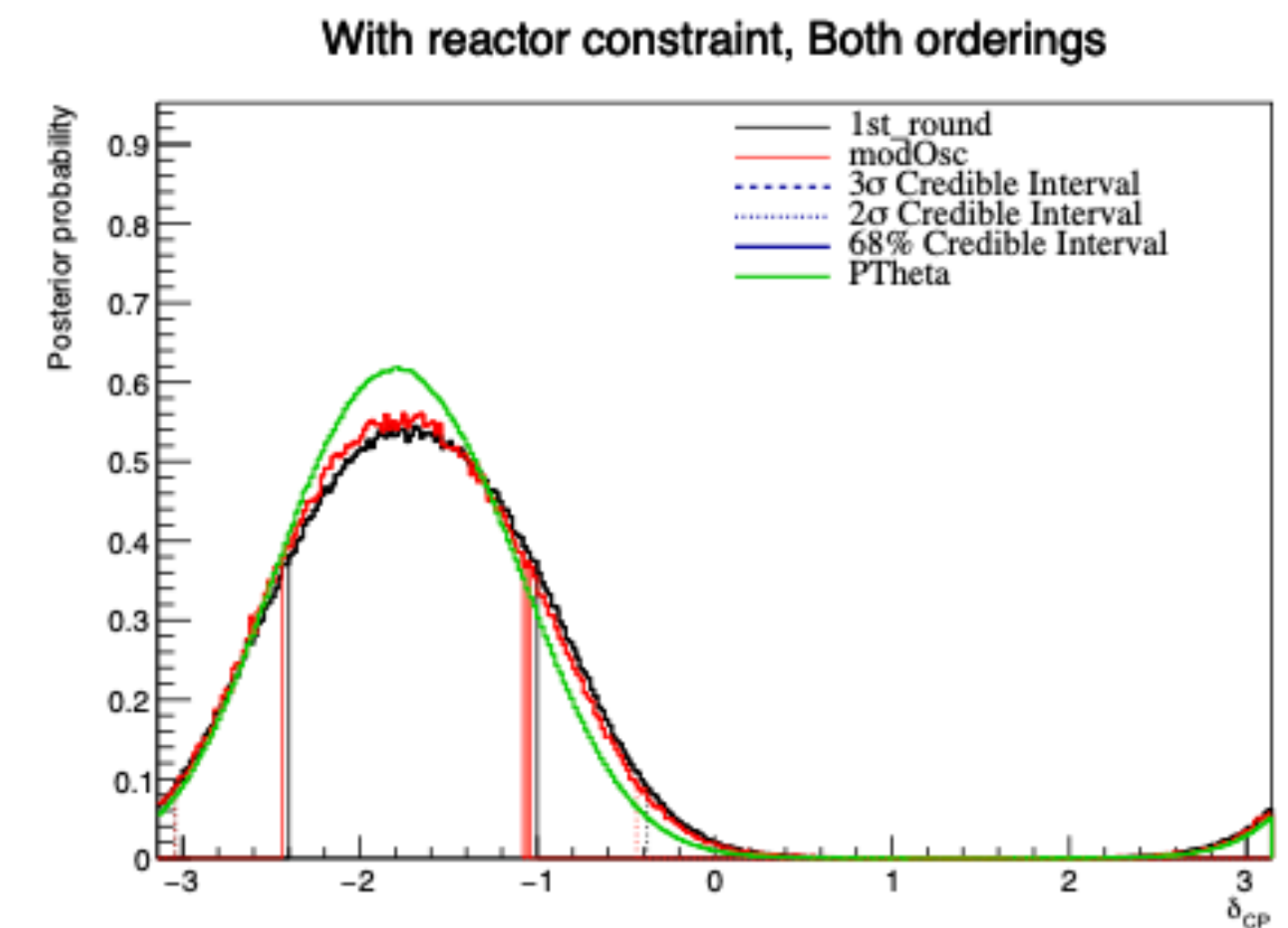
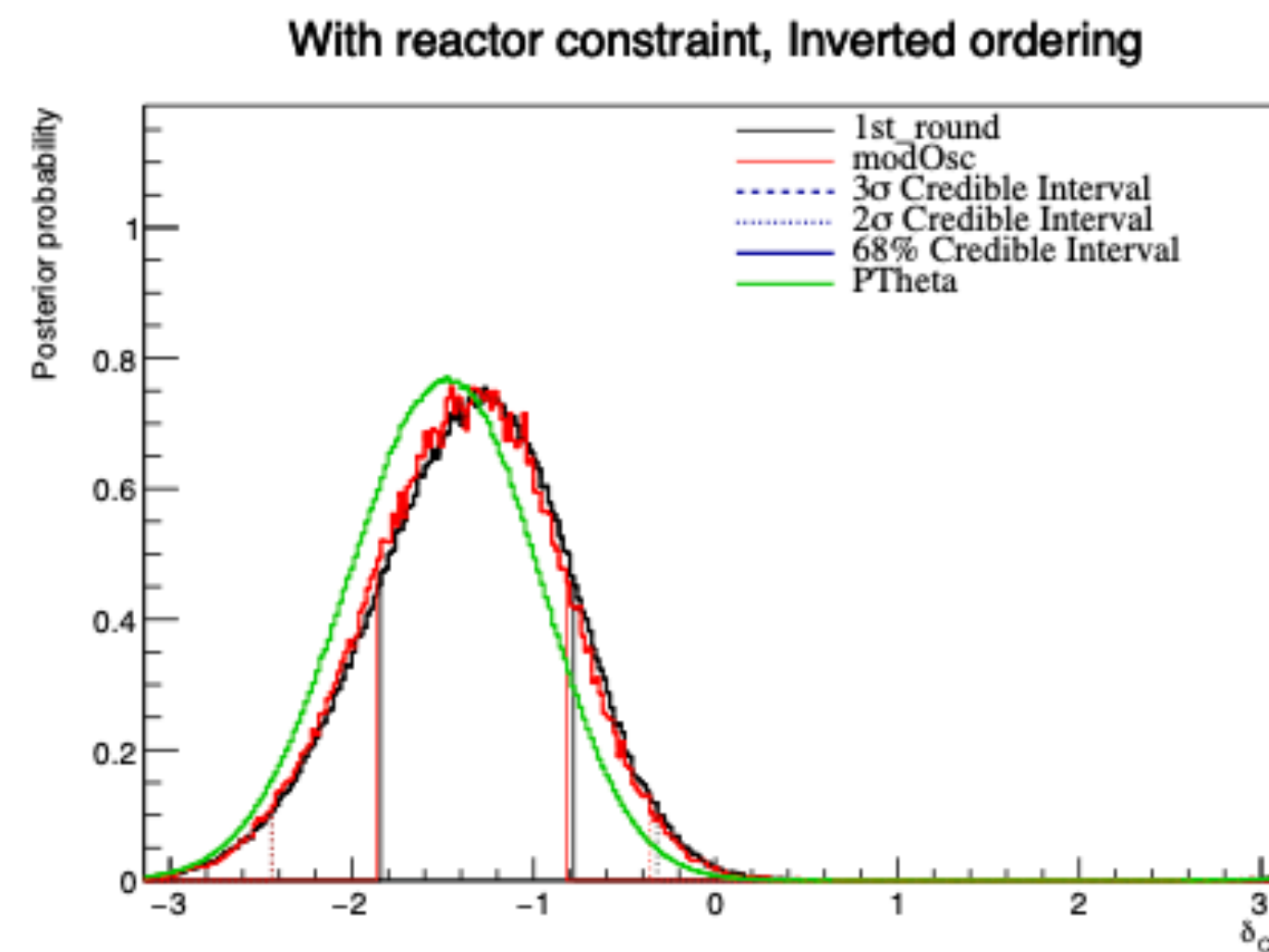
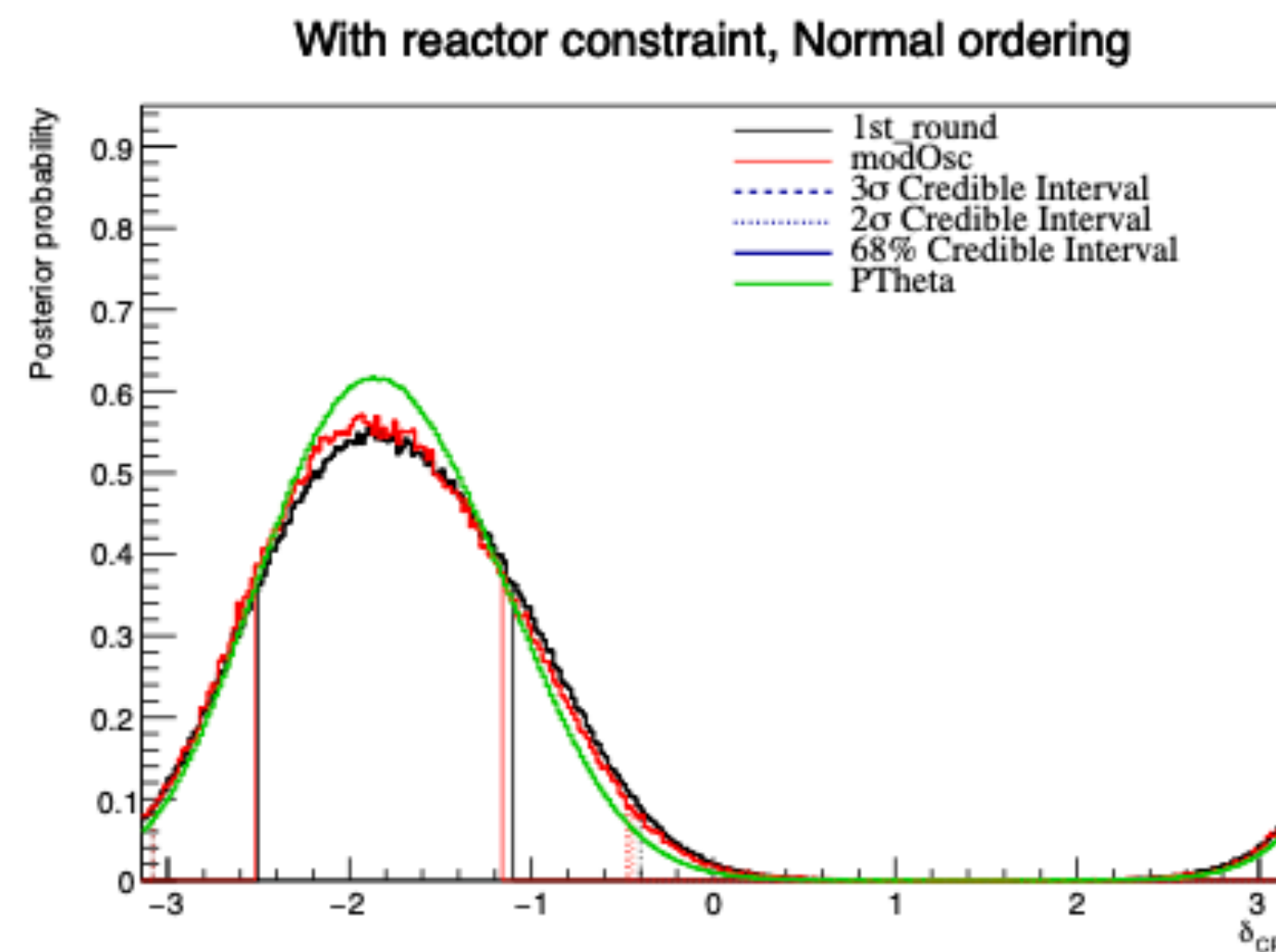
Warsaw Group Meeting, 28.09.2026

Reminder of the p-theta work

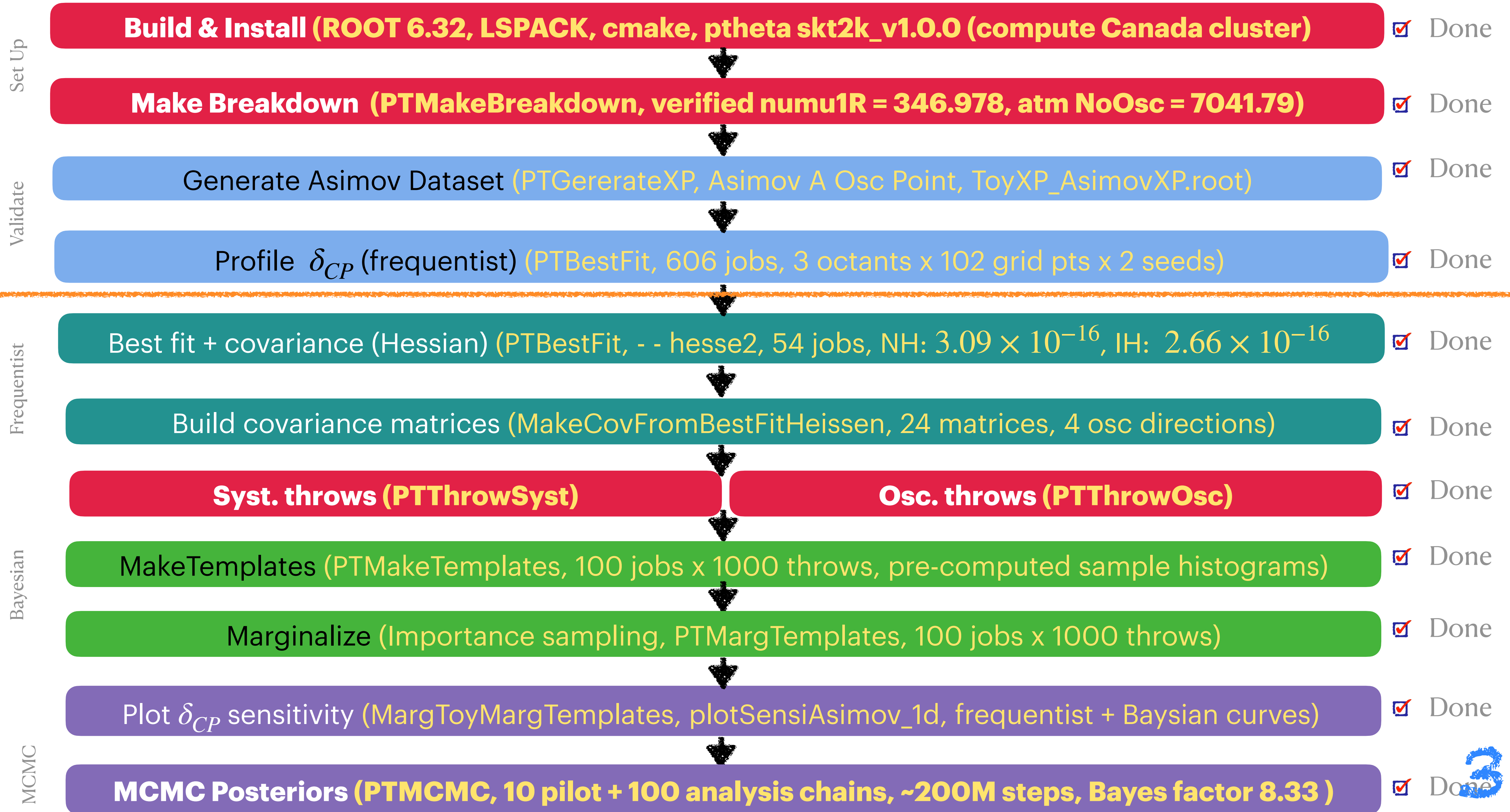
Momentum scale: do correlations between beam and atmospheric matter?

Problem to understand:

- * In ptheta and osc3++, the beam and atm p-scale parameters were fully correlated
- * In the mach3 analysis, they were uncorrelated
- * There was a shift in data fit between ptheta and mach3



Workflow



Work flow...

Two steps were not done yet:

- * Real data fit for δCP was skipped

The data fit requires the official SK+T2K data files. I have access to the Asimov files but not yet to the official data input files on sukapo1. This is the next practical step — I need to locate the correct data path that TN459 used

- * MCMC is partial (Asimov only, no multicanonical)

TN459 used the multicanonical method to properly sample the IH region. This means, in my case IH posterior is undersampled and Bayes factor (8.33) is less accurate than TN459's

Frequentist δ_{CP} Profile Fit

The frequentist δ_{CP} profile fit **scans δ_{CP} across its full range**, and at each value finds the best possible fit to the data by **minimising over all other parameters** — producing a $\Delta\chi^2$ curve that shows which values of δ_{CP} the data supports and which it rules out

Frequentist δ_{CP} Profile Fit

What was done:

- Scanned δ_{CP} grid using **PTBestFit** (MINUIT minimization)
- At each grid point: fix δ_{CP} and mass ordering, minimize over all ~200 nuisance parameters
- Profile likelihood ratio: $\Delta\chi^2(\delta_{CP}) = -2 \ln[L(\delta_{CP})/L(\delta_{CP},\text{best})]$

Compute Canada resources:

Partition: def-blairt2k
Wall time per job: 1 hour
Memory: 4 GB/job
All 606 jobs completed, **0 missing output files**

Grid structure:

Parameter	Values
δ_{CP} grid points	51 points $\times$ 2 mass orderings = 102
θ_{23} octants	3 (lower, upper, free)
Seeds per point	2 (different random initializations)
Total jobs	$3 \times 102 \times 2 = 606$

Post processing:

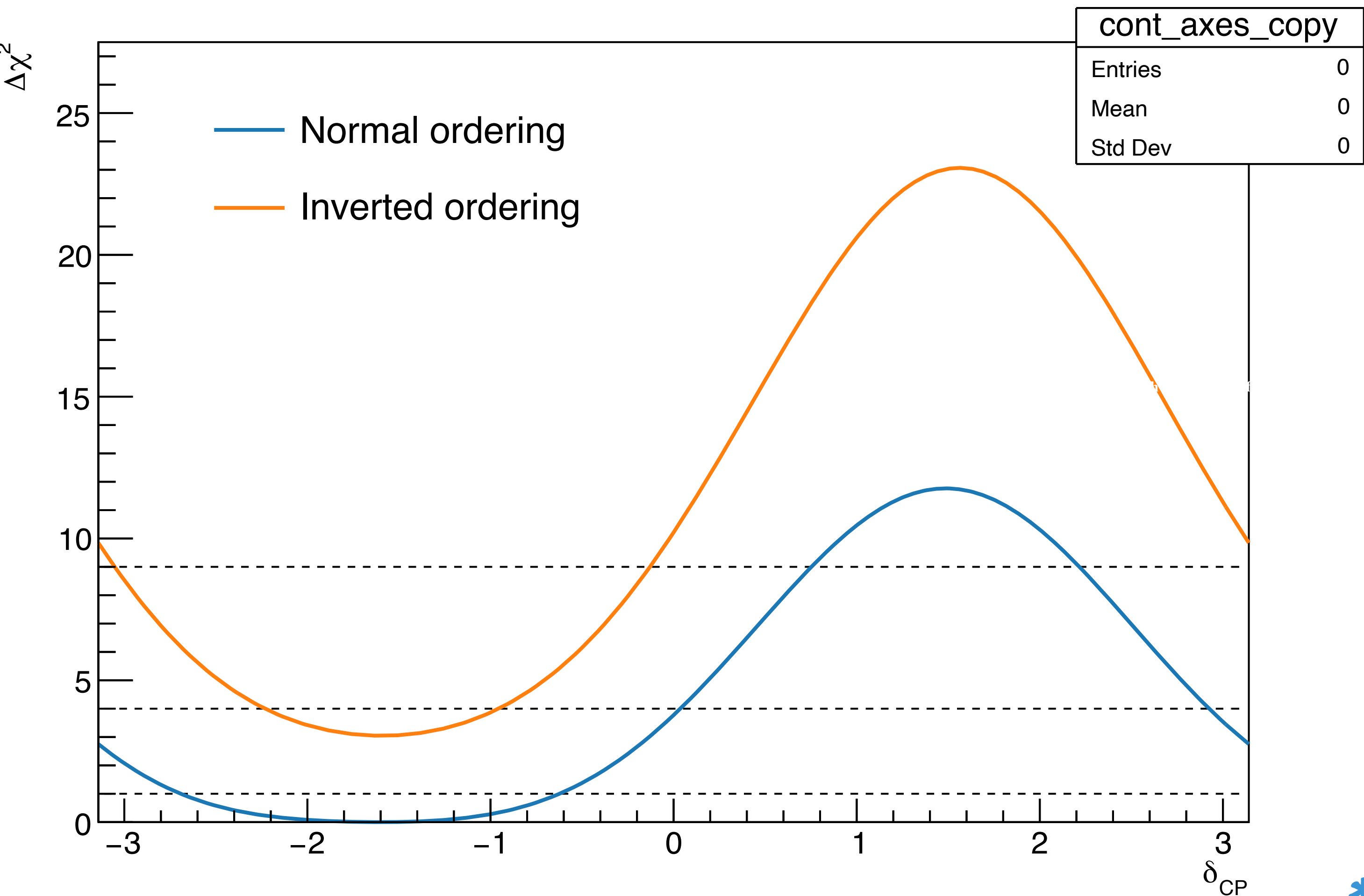
MergeToyBestFit.cxx — merge 600 root files
PlotSensiAsimov_1D_parallel.C — compute profile per grid point
PlotSensiAsimov_1D_combined.C — produce final pdf plot

Note — bug fixed:

PlotSensiAsimov_1D_parallel.C **line 196** dereferences MargTemplatePerToy without null check when reading BestFit output — fixed by adding if (MargTemplatePerToy) guard



Frequentist δ_{CP} Profile Fit



Blue curve = Normal Ordering (NH) — minimum at $\delta_{CP} \approx -1.60$ rad, $\Delta\chi^2 = 0$

Orange curve = Inverted Ordering (IH) — offset upward, minimum $\Delta\chi^2 = 3.05 \rightarrow$ NH preferred

Dashed lines = $1\sigma, 2\sigma, 3\sigma$ thresholds ($\Delta\chi^2 = 1, 4, 9$)

Observable	From my test (Asimov)	TN459 (data fit)
NH best-fit δ_{CP}	-1.603 rad	$-1.76^{+0.60}_{-0.74}$ rad
IH $\Delta\chi^2$ at minimum	3.05	~ 3.5

- * $\Delta\chi^2 < 1$: within $1\sigma \rightarrow$ strongly consistent with data
- * $\Delta\chi^2 < 4$: within $2\sigma \rightarrow$ still consistent with data
- * $\Delta\chi^2 > 9$: outside $3\sigma \rightarrow$ strongly disfavoured

Next step: full posterior probability distribution over δ_{CP} , properly accounting for all nuisance parameters

Bayesian analysis

For the Bayesian analysis, p-theta uses **importance sampling** rather than **direct MCMC over all parameters**. Importance sampling uses the **best-fit point and Hessian** from the **frequentist step** to define a Gaussian proposal distribution, then **reweights the samples** to get the correct posterior. This runs in four steps — **systematic throws**, **oscillation throws**, **making templates**, and **marginalizing** — totalling 200 jobs, all completed

A Bayesian **can** say "there is a 90% probability that δCP is between -2.5 and -0.9 rad" — this is a statement about how certain we are, given the data and our prior knowledge



Procedure

From frequentist test already have:

- * The best-fit values of all parameters (from MINUIT)
- * The **Hessian matrix** — the matrix of second derivatives of χ^2 at the minimum

1. **PTThrowSyst (systematic throws)**: Draw 100,000 samples of the ~200 systematic nuisance parameters from the Gaussian proposal

These are 100k "hypotheses" about what the systematic parameters might be

2. **PTThrowOsc (oscillation throws)**: Separately draw 100,000 samples of the oscillation parameters (δCP , $\sin^2\theta_{23}$, Δm^2_{32}) from their prior distributions

These cover the full range of possible oscillation parameter values you want to probe

Procedure

3. PTMakeTemplates (100 jobs $\times$ 1000 throws): Separately draw 100,000 samples of the oscillation parameters (δCP , $\sin^2\theta_{23}$, Δm^2_{32}) from their prior distributions

For each combination of systematic throws and oscillation throws, compute the **predicted event histogram** — the template — for all samples

This is the expensive step. For each of the 100k throws:

- Apply the systematic parameters to the MC (reweighting)
- Compute the predicted bin counts for all atmospheric and beam samples
- Store the resulting histogram

4. PTMargTemplates (100 jobs $\times$ 1000 throws): compute the importance weights and marginalise

For each throw i with parameters $(\theta_{\text{syst}}^i, \theta_{\text{osc}}^i)$:

- Look up the precomputed template (from step 3)
- Compare it to the data (or Asimov) $\rightarrow$ compute the likelihood L^i
- Compute the importance weight: $w^i = \frac{L^i \cdot \pi(\theta^i)}{Q(\theta^i)}$, π : prior, Q : Gaussian proposal
- For each δ_{CP} grid point, sum up the weights of all throws at that δ_{CP} value

A weighted sum over all nuisance parameter throws — this **marginalises** over the nuisance parameters and gives the posterior $P(\delta\text{CP} \mid \text{data})$

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Bayesian Analysis — Importance Sampling Pipeline

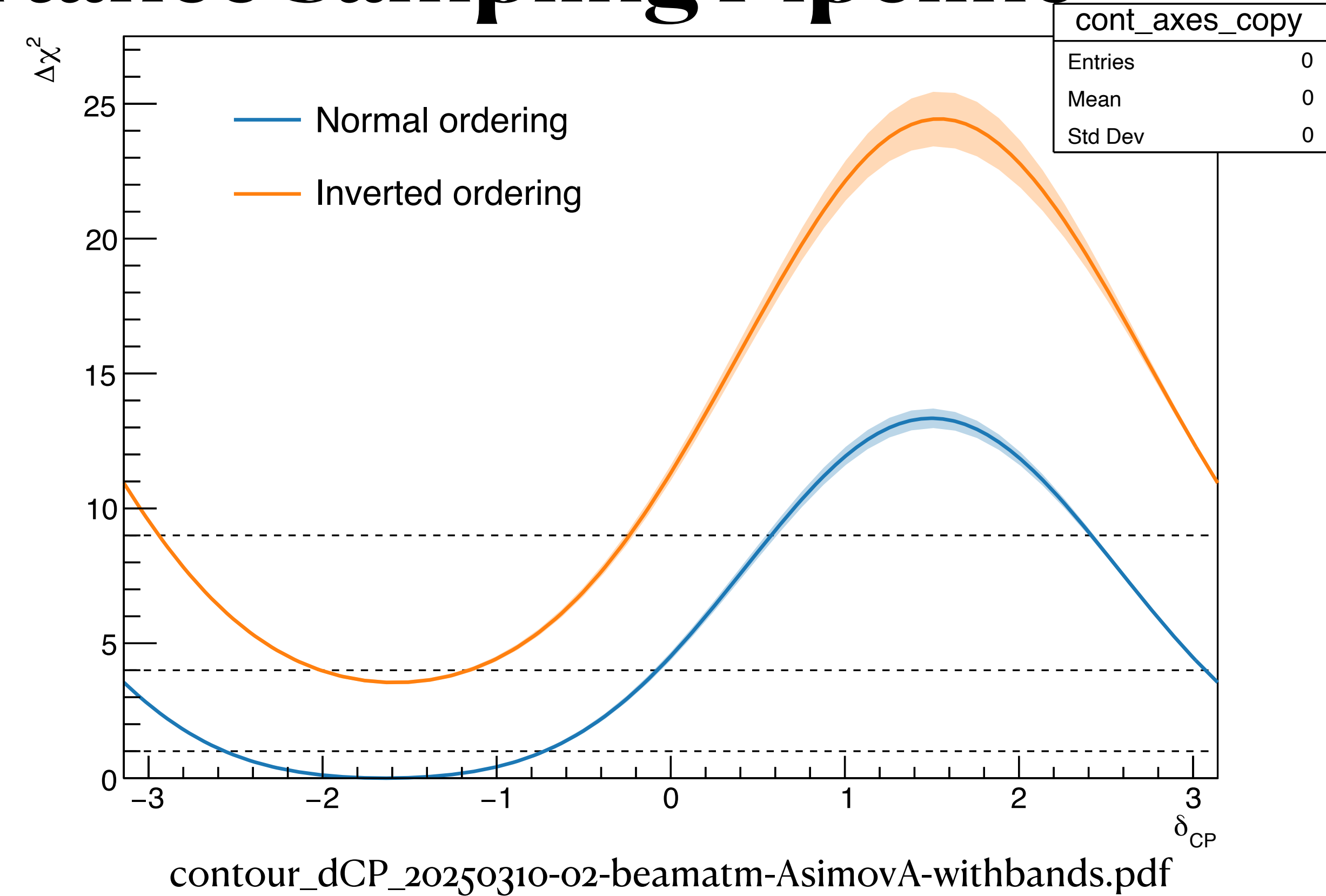
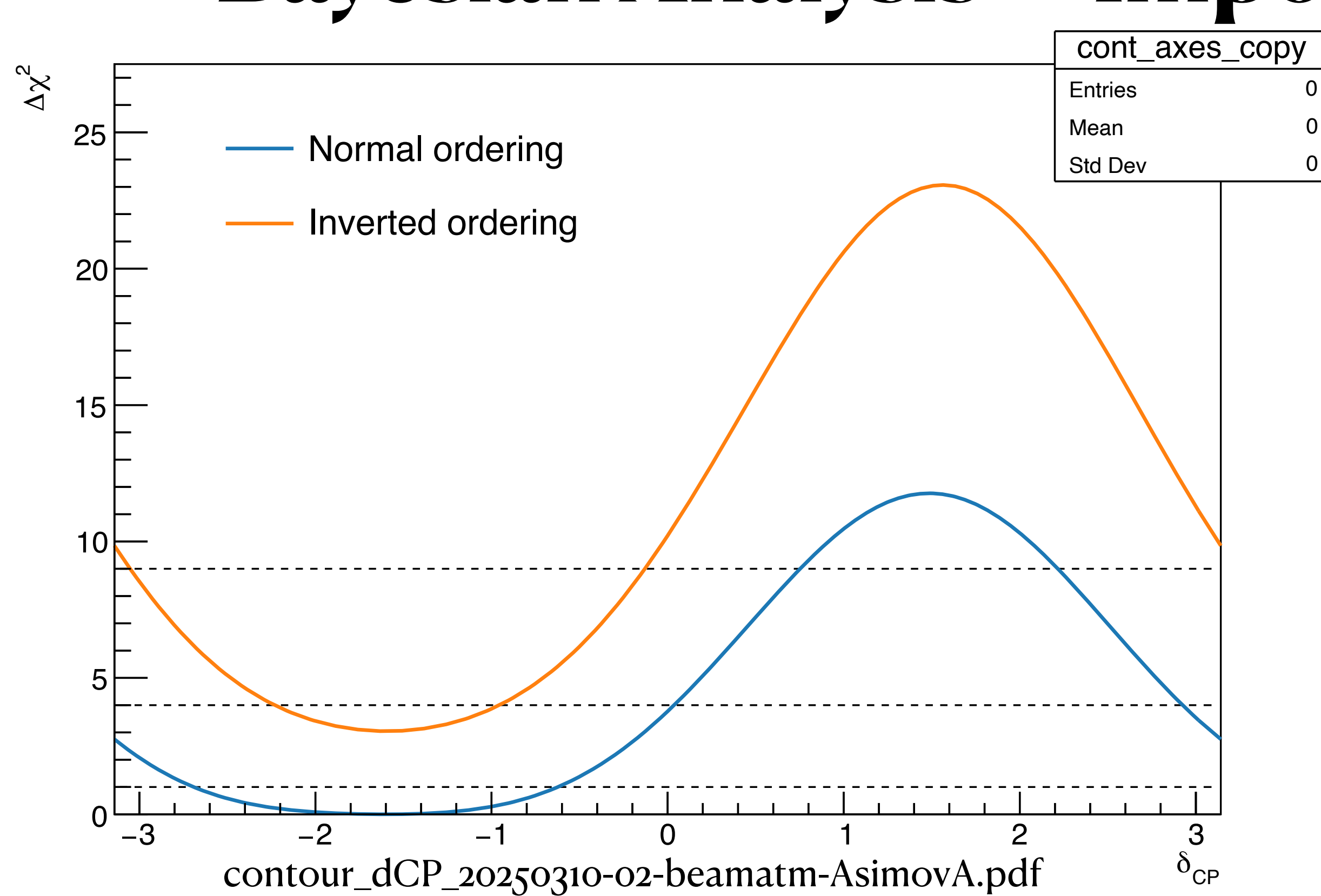
Why importance sampling?

- Full marginalization over ~ 200 nuisance parameters requires integrating the posterior
- Direct MCMC over all parameters is expensive — importance sampling uses a **Gaussian approximation** as a proposal, then reweights
- p-theta uses best-fit + Hessian (Steps 5–6) to define the proposal distribution

set-up:

Step	Tool	Jobs	Output
Systematic throws	PTThrowSyst	1 job	100k syst. parameter samples · 164 MB ☒ Done
Oscillation throws	PTThrowOsc	1 job	100k osc. parameter samples · 8.1 MB ☒ Done
Make templates	PTMakeTemplates	100 jobs	Pre-computed histograms per throw ☒ Done
Marginalize	PTMargTemplates	100 jobs	$\Delta\chi^2$ profile per δ_{CP} grid point ☒ Done

Bayesian Analysis — Importance Sampling Pipeline



Blue curve = NH — minimum at $\delta_{CP} \approx -1.64$ rad

Orange curve = IH — consistently higher, $\Delta\chi^2 \approx 3.5$ at minimum

Shaded bands = marginalization statistical uncertainty from finite 100k throws — bands are narrow → **sampling is well converged**

Dashed lines = 1σ , 2σ , 3σ thresholds

Profiling **fixes nuisance parameters at their best-fit value** for each δ_{CP} grid point. Marginalization integrates over all nuisance parameter values. When the posterior is not perfectly Gaussian in the nuisance parameters, these two procedures give slightly different results

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Bayesian Analysis — Importance Sampling Pipeline

Observable	Frequentist (profiling)	Bayesian (IS)
Method	Minimize over nuisance params	Integrate over nuisance params
NH best-fit δ_{CP}	−1.603 rad	−1.643 rad
IH $\Delta\chi^2$ at minimum	3.05	~3.5
Error bands	None	Yes (IS statistical uncertainty)
TN459 reference δ_{CP}	−1.76 rad	−1.87 rad

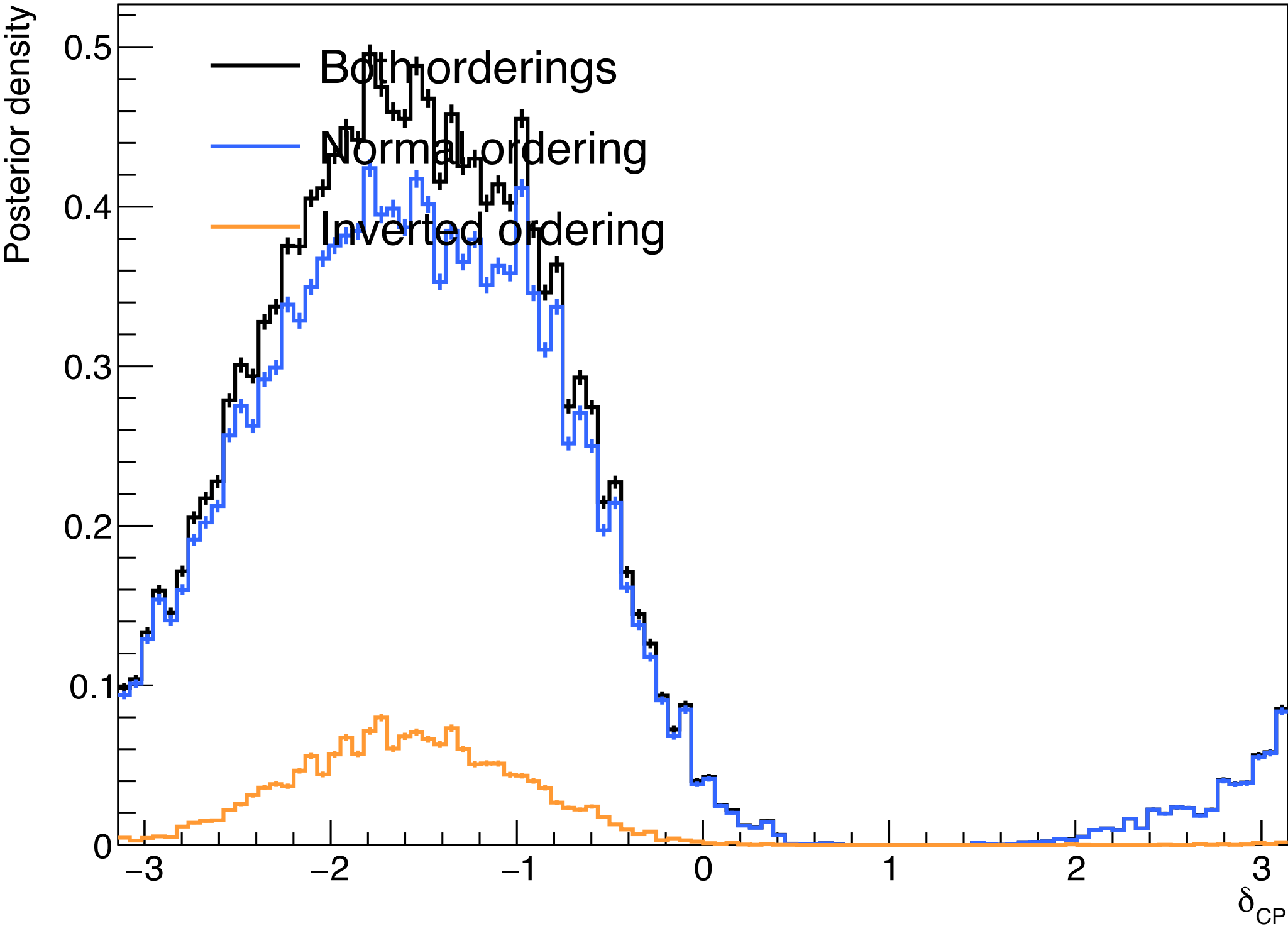
The ~0.04 rad difference between frequentist and Bayesian best-fits reflects — profiling and marginalization treat nuisance parameters differently

Both are consistent with TN459 in shape and mass ordering preference

Bayesian Analysis — MCMC Configuration

Why MCMC?

- Importance sampling gives the δ_{CP} profile but not the full posterior distribution
- MCMC samples the **full joint posterior** $P(\delta_{\text{CP}}, \sin^2\theta_{23}, \Delta m^2_{32}, \text{MO} \mid \text{data})$ directly
- Enables credible intervals, Bayes factors, and posterior predictive checks

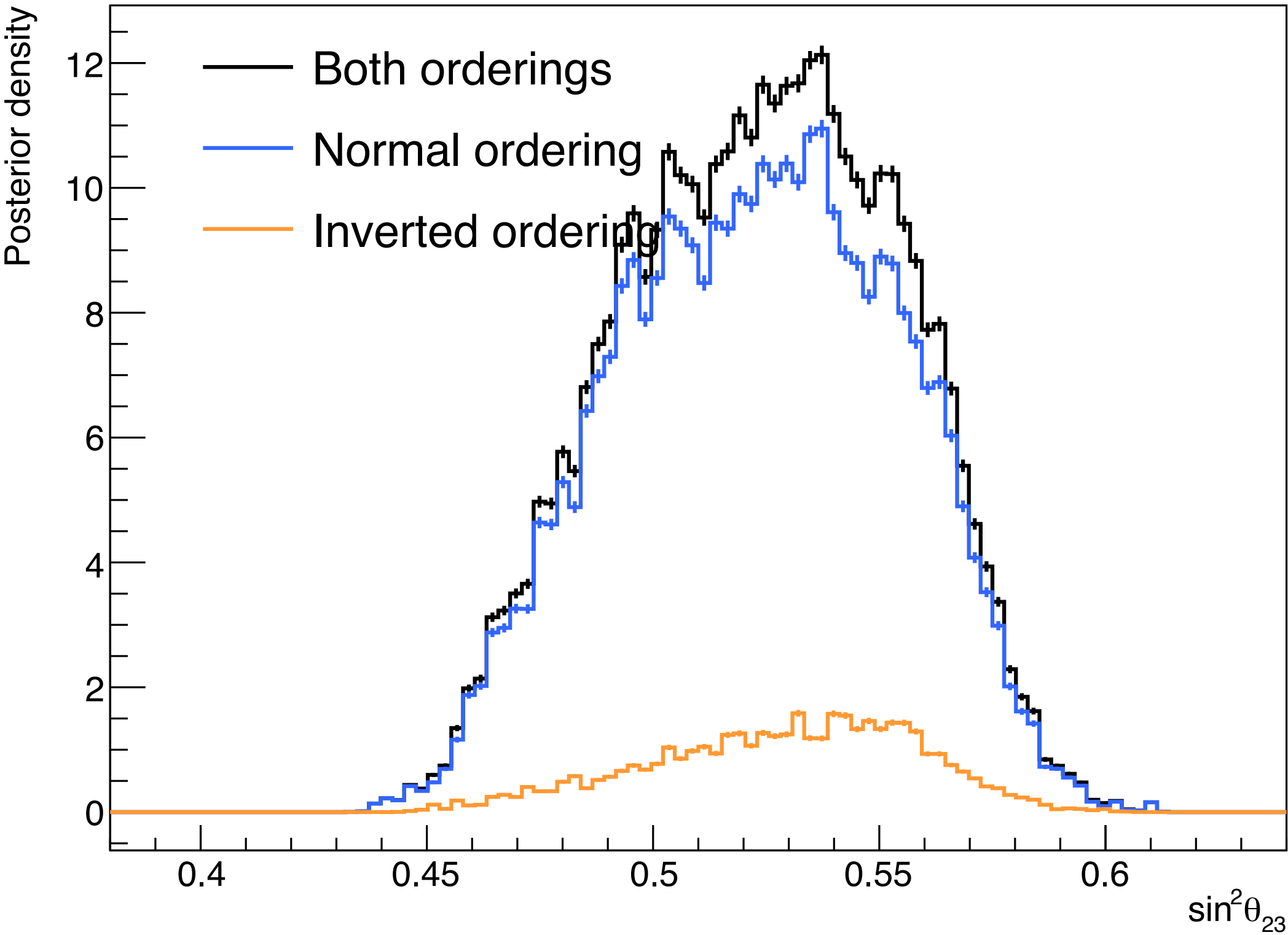


- **Black curve = both orderings combined**
- **Blue curve = Normal Ordering (NH)** — peak near $\delta_{\text{CP}} \approx -1.6$ rad
- **Orange curve = Inverted Ordering (IH)** — broader, lower peak, subdominant
- **Y-axis** = posterior density (normalized so total area = 1)
- NH dominates — consistent with Bayes factor favouring Normal Ordering

Observable	Our result (Asimov)	TN459 (data fit)
NH most probable δ_{CP}	~ -1.6 rad	$-1.87^{+0.67}_{-0.59}$ rad
IH most probable δ_{CP}	~ -1.6 rad	$-1.48^{+0.50}_{-0.53}$ rad
2σ excludes CP conservation?	Consistent with TN459	Yes (flat δ_{CP} prior)
Shape agreement	✓ Good	Reference

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MCMC Result — $\sin^2\theta_{23}$ and Δm^2_{32} Posteriors



$\sin^2\theta_{23}$:

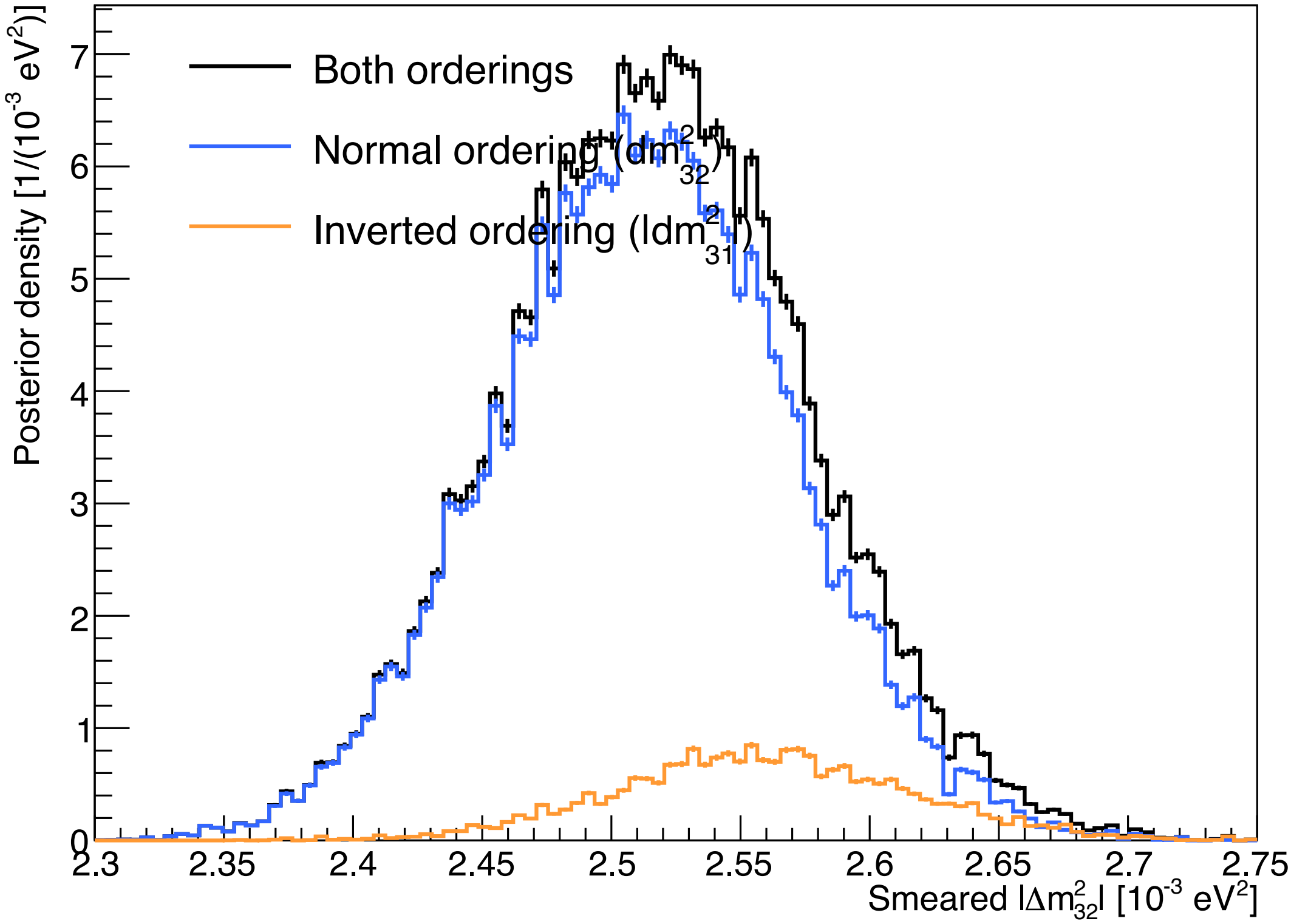
Blue (NH): peak around 0.55, upper octant preference

Orange (IH): similar shape, slightly shifted

Black (combined): broad peak spanning upper octant

AsimovA truth: $\sin^2\theta_{23} = 0.528$ — posterior peak consistent ✓

Observable	Our result (Asimov)	TN459 (data fit)
NH peak $\sin^2\theta_{23}$	~ 0.55	$0.549 [0.459, 0.505] \cap [0.521, 0.568]$
IH peak $\sin^2\theta_{23}$	~ 0.56	$0.558^{+0.057}_{-0.050}$



$|\Delta m^2_{32}|$ (smeared, IH-corrected):

Blue (NH): sharp peak at $\sim 2.51 \times 10^{-3} \text{ eV}^2$

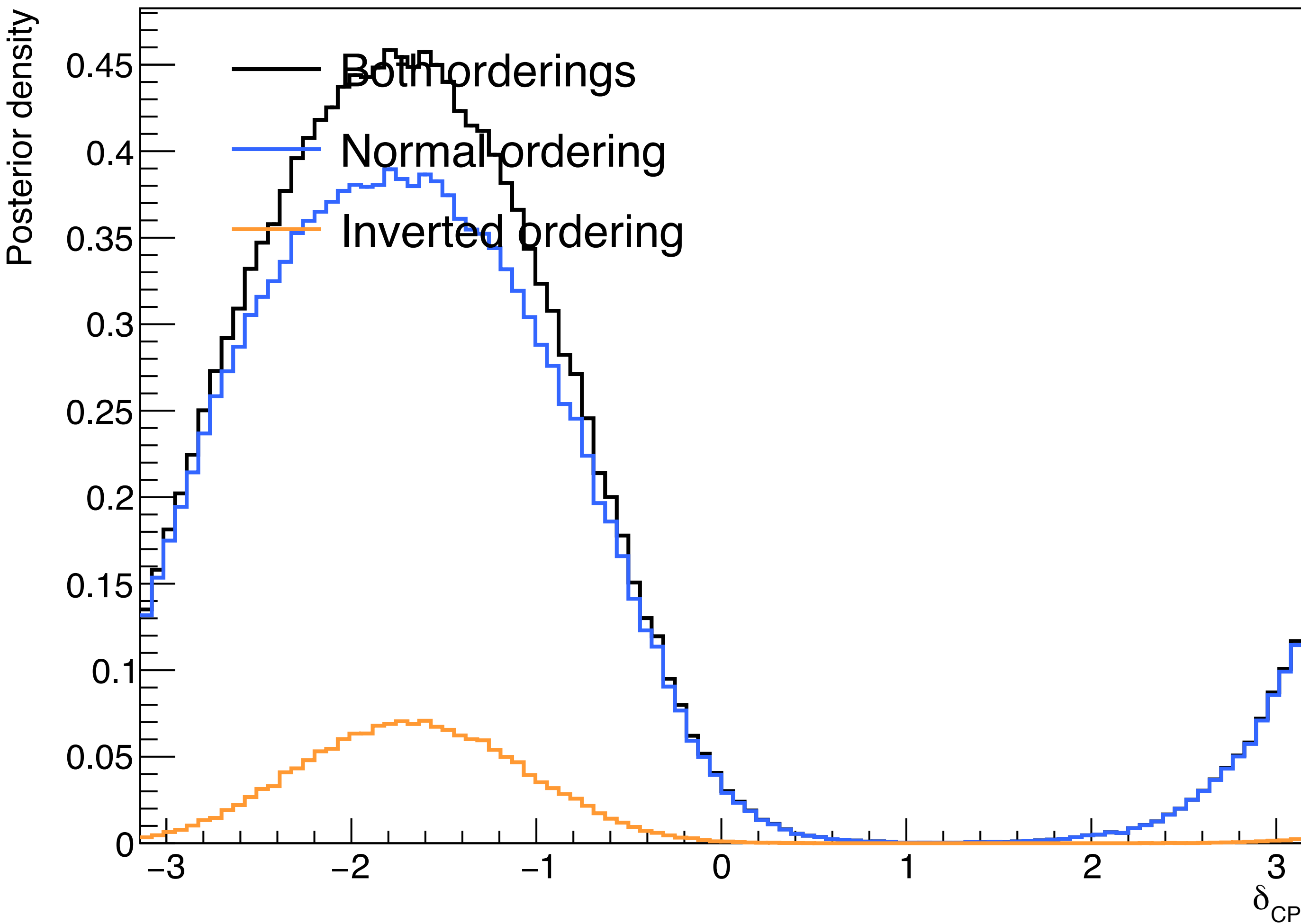
Orange (IH): peak at $\sim 2.49 \times 10^{-3} \text{ eV}^2$ (using $|\Delta m^2_{31}| = \Delta m^2_{32} + \Delta m^2_{21}$)

Δm^2_{32} smearing of $3.6 \times 10^{-5} \text{ eV}^2$ applied (as in TN459) via yp_core proxy variable

AsimovA truth: $\Delta m^2_{32} = 2.509 \times 10^{-3} \text{ eV}^2$ — posterior peak consistent ✓

Observable	Our result (Asimov)	TN459 (data fit)
NH peak Δm^2_{32}	$\sim 2.51 \times 10^{-3} \text{ eV}^2$	$2.511^{+0.060}_{-0.058} \times 10^{-3} \text{ eV}^2$
IH peak $ \Delta m^2_{31} $	$\sim 2.49 \times 10^{-3} \text{ eV}^2$	$2.483^{+0.024}_{-0.060} \times 10^{-3} \text{ eV}^2$

MCMC Result — Mass Ordering Bayes Factor



Black curve = both orderings combined, peak near $\delta_{CP} \approx -1.6$ rad
Blue (No = Normal Ordering) = dominant, clear peak
Orange (Inv = Inverted Ordering) = much smaller area

Observable	Our result (Asimov)	TN459 (data fit)
Total MCMC steps	~5.35M	200M
NH steps	4,774,675 (89.3%)	~91%
IH steps	573,325 (10.7%)	~9%
Bayes factor B(NH/IH)	8.328	8.98 ± 0.06
Significance	~1.60σ	1.64σ

What does a Bayes factor of 8.33 mean physically?
It means the data is 8.33 times more likely under Normal Ordering than under Inverted Ordering

Bayes factor = $\frac{\text{number of MCMC steps spent in NH}}{\text{number of MCMC steps spent in IH}}$

What done till now...

Step	Status	Notes
Make breakdown	✓ Full	Exact numerical match
Make Asimov	✓ Full	Identical output
Fit Asimov for δ_{CP}	✓ Full	Same jobs, same plots
Fit data for δ_{CP}	✗ Skipped	Real data fit not run
Get best-fit + cov	✓ Full	Volume reductions match
Marginalize Asimov (IS)	✓ Full	Slightly larger sensitivity !?
Run MCMC	⚠ Partial	Asimov + no multicanonical + own pilot chains

Next step ...

- * **Reproducing the Data Fit:**

- * Waiting for some informations:

- * Are the real data files already accessible on sukap01 at the path the TN459 config expects?
 - * Is there an existing data fit submission script ready to run, or do I need to modify the Asimov one?
 - * Is there a reference output from the official TN459 data fit chain that I can compare against?

- * **Constructs the response function by correctly accounting for inter-sample migrations**

- * Procedure follows these steps:

- * Take the raw MC events
 - * Scale the reconstructed momentum by the parameter value
 - * Redo the full atmospheric event selection with the scaled quantities
 - * Bin the events into their **post-scaling sample** using the scaled momentum

Three main ingredients

Likelihood — $P(\text{data} \mid \delta\text{CP})$:

How probable is the observed data if δCP takes this value? This comes from comparing predicted event counts to observed counts. This is the same likelihood used in the frequentist fit.

Prior — $P(\delta\text{CP})$:

What did we believe about δCP before seeing this data? P-theta uses a flat prior — all values of δCP are equally likely before seeing data. This is the most neutral choice.

Posterior — $P(\delta\text{CP} \mid \text{data})$:

What do we believe about δCP after seeing the data? This is the Bayesian result — a full probability distribution over δCP .

What is frequentist δCP profile fit?

We want to measure δ_{CP} (parameters that tells us about matter/anti-matter asymmetry), it could be in the range of $-\pi$ to π . The question is which value of δ_{CP} are consistent with data, and which are ruled out.

Basic idea:

For any given value of δ_{CP} , we can ask: How well does the data fit if the δ_{CP} is fixed at this value?

We scan δ_{CP} across its full range — trying different values — and at each value we measure how well the model fits the data. Values where the fit is good are **allowed**. Values where the fit is bad are **excluded**.

What profile means:

There are other fit parameters besides δ_{CP} . We can't just fixed everything and vary only δ_{CP} . So at each δ_{CP} grid point, minimize over all other parameters and find the best possible value of other parameters, given this fix δ_{CP}

Why Frequentist:

Frequentist based on likelihood - how often you would get data at least as extreme as what you observed , if a given δ_{CP} were true. It gives confidence intervals, not probability statements.

Importance Sampling

After the frequentist fit gives us the δ CP profile, we want the **Bayesian result** — a full posterior probability distribution over δ CP, properly accounting for all nuisance parameters

To get this, we need to **integrate** — add up the probability over all possible values of all nuisance parameters:

$$P(\delta_{CP} | data) = \int P(\delta_{CP}, \theta | data) d\theta$$

The most straightforward approach: run MCMC over all 200+ parameters simultaneously. At each step, propose new values for everything, accept or reject based on the likelihood.

The problem: with 200 parameters, the chain takes an enormous number of steps to explore the full space and converge. TN459 uses 200M steps even for the MCMC posterior — and that is after importance sampling has already been done for the δ CP profile. Running direct MCMC for the profile would be far more expensive.

What is the multicanonical method?

The multicanonical method is a **technique that biases the MCMC to explore regions of low probability** — specifically the IH region — more thoroughly, then corrects for the bias. Without it, the chain spends very little time in IH because it is disfavoured, making the Bayes factor estimate noisy. I did not use it because it requires additional configuration and computing time. This is noted as partial in the status table